Shih-Ni Prim
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Dear hiring manager,

I am writing to express my interest in the position of data scientist, additive manufacturing at LEGO. Currently a third-year PhD student in statistics at North Carolina State University in the United States, I am experienced with developing novel statistical and machine learning methods, programming with R and Python, conducting exploratory data analysis, creating visualization to understand data and to communicate findings, and presenting my research in written and verbal formats. With my skills and experience, I am ready to take on the challenges and exciting opportunities this position offers.

During my time in the master's and doctoral programs in statistics, I have worked on a wide variety of research projects about data analysis and machine learning. The research projects for my courses included tensor decomposition, using machine learning methods (random forests, XGBoost, Gaussian Process, neural networks) to predict critical temperatures for semiconductors, causal inference for multiple exposures and multiple outcomes, transfer learning, etc. As a research assistant, I worked with a biological and agricultural engineering professor and an algal physiology and ecology professor to analyze time series data to find factors that contribute to harmful algal bloom. The project required extensive data cleaning and preprocessing, data analysis and modeling, visualization, and writing of journal publication. Although I have not had direct experience with additive manufacturing, my experience in data analysis, statistical and machine learning methods, and collaboration with researchers outside statistics has prepared me for working with engineers and technicians at LEGO and contributing to process optimization, quality assurance, and predictive modeling.

Besides assisting with others' research, my own doctoral research requires creativity and critical thinking to solve problems. My research includes two very different projects. One project develops a fully Bayesian, spectral method to analyze the effects of multiple exposures with the application of analyzing health burdens of per- and polyfluoroalkyl substances (PFAS). A second project designs an active learning method to optimize how to best sample points for a surrogate model (using deep Gaussian Process) for computer experiments with the application of finding best conditions for stable and efficient flights. The wild differences between the two projects in terms of data and collaborators—geospatial data and epidemiologists as collaborators versus fluid dynamic simulator data and aerospace engineers as collaborators—have helped me gain breadth in knowledge about machine learning modeling. For both projects, my advisor, mentors, and I develop novel statistical and machine learning methods. Many problems inevitably occur during the process, and each time we carefully test the algorithms to understand what works and what does not work. When we apply the methods on real-world data, more problems often occur, and we sometime have to go back to the drawing board. The research process is challenging but deeply rewarding. I found myself truly enjoying the creative side of constructing machine learning models and decided to find a career that provides room for creativity and challenges, both of which this data scientist position offers.

LEGO has been the most loved toys for my two children, and I believe that LEGO helps children learn so much about engineering and creating, with or without them realizing it. For this reason, I am eager to join the company. I understand that, as a foreigner, I do not have the advantage an EU citizen has. I hope that this cover letter conveys my excitement about and qualifications for this position. Should I be selected for this role, I plan to start learning Danish before relocation and finish my current PhD degree remotely outside work hours. Thank you for your time and consideration, and I look forward to hearing from you.

Sincerely,

Shih-Ni Prim